Digital Signal Processing **HK261** **CO2036**

DIGITAL SIGNAL PROCESSING LAB REPORT

Lab 3 – Script Functions for Discrete-Time Signals

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Overview

This report presents the results of Lab 3 of the Digital Signal Processing laboratory. In this lab, the basic time-domain operations on discrete-time signals (delay, advance, folding, addition, multiplication and convolution) are implemented as reusable **Scilab script functions**. Every function receives a signal as a pair (`vector`, `origin position`), returns the resulting signal in the same format and graphically displays the input(s) and output in a single figure. The report also covers the additional exercises 2.1, 2.2, 2.4 and 2.6 of the reference document, both theoretically and practically on Scilab.

Table 1: Content of the report

Part	Content	Section
Exercise 1	`delay` – $y(n) = x(n-k)$	4.1
Exercise 2	`advance` – $y(n) = x(n+k)$	4.2
Exercise 3	`fold` – $y(n) = x(-n)$	4.3
Exercise 4	`add` – $y(n) = x_1(n) + x_2(n)$	4.4
Exercise 5	`multi` – $y(n) = x_1(n) \cdot x_2(n)$	4.5
Exercise 6	`convolution` – $y(n) = x(n) * h(n)$	4.6
Additional 2.1	Fold / delay of a signal	4.7
Additional 2.2	Sketching transformed signals	4.7
Additional 2.4	Even – odd decomposition	4.7
Additional 2.6	Time-invariance of systems	4.7

1 Introduction

Lab 1 and Lab 2 introduced Scilab and the elementary discrete-time signals, where each operation was written directly in the console for a particular signal. In Lab 3, these operations are generalized into script functions so that they can be applied to *any* finite-length sequence. The key difficulty is that a Scilab vector has no notion of the time index n: its elements are always indexed from 1. Therefore, each signal is represented by a vector `xn` together with an integer `xorigin` which stores the position of the sample $x(0)$ inside the vector. The time index of the element at position i is then

$$n_i = i - \texttt{xorigin}, \qquad i = 1, 2, \ldots, \texttt{length(xn)}. \tag{1}$$

2 Objectives

- Review the lecture notes on discrete-time signals and systems.
- Implement script-based manipulation functions in Scilab for discrete-time signals.
- Verify each function with an example and capture the output (console and figure) as evidence.
- Solve the additional exercises 2.1, 2.2, 2.4 and 2.6 both theoretically and on Scilab.

3 Methodology

3.1 Scilab script functions

A script function in Scilab is declared with the following format:

```
function [rv1, rv2, ..., rvn] = Function_Name(pv1, pv2, ..., pvn)
    // function body
endfunction
```

where `rv1, rv2, ...` are the return values and `pv1, pv2, ...` are the input parameters. Each function is stored in a `.sci` file, loaded with `exec('file.sci', -1)` and then called from the console.

3.2 Common conventions

All functions in this lab follow the same conventions:

- A signal is passed as a pair (`xn`, `xorigin`) and returned as (`yn`, `yorigin`).
- The time index vector is built from (1): `n = (1:length(xn)) - xorigin`.
- Signals are drawn with `plot2d3` (stems) plus circle markers, inside `subplots` of a single figure. All subplots share the same axis bounds so that the shift of the samples can be compared directly.

4 Practical Contents

4.1 Exercise 1 – Delay

Question: Write the function `[yn, yorigin] = delay(xn, xorigin, k)` that performs the delay operation $y(n) = x(n-k)$, where $k > 0$. $x(n)$ and $y(n)$ are graphically displayed in the same figure.

For example, $y(n) = x(n-1)$ with $x(n) = \{1, -2, \underset{\uparrow}{3}, 6\}$ is computed by `[yn, yorigin] = delay([1, -2, 3, 6], 3, 1)` and the output is `yn = [1, -2, 3, 6]`, `yorigin = 2`.

Explanation:
When a discrete-time signal $x(n)$ is delayed by $k > 0$ samples to produce $y(n) = x(n-k)$, the entire signal shifts to the right along the time axis. In an array-based representation, the sample values themselves do not need to be reordered; instead, the index pointing to the "time zero" origin shifts to the left by k positions.

For example, consider $x(n) = \{1, -2, 3, 6\}$ with the origin at the 3rd element ($x(0) = 3$). If we delay this by $k = 1$ to get $y(n) = x(n-1)$, the value at $n = 1$ becomes $y(1) = x(0) = 3$. The value at $n = 0$ becomes $y(0) = x(-1) = -2$. Thus, the new origin for $y(n)$ is now at the 2nd element of the array. If shifting left causes the origin index to fall below 1 (e.g., $k \geq$ xorigin), the array must be zero-padded on the left to maintain a valid array index for the origin.

Result:

Console output

```
yn =
1.  -2.  3.  6.
yorigin =
2.
```

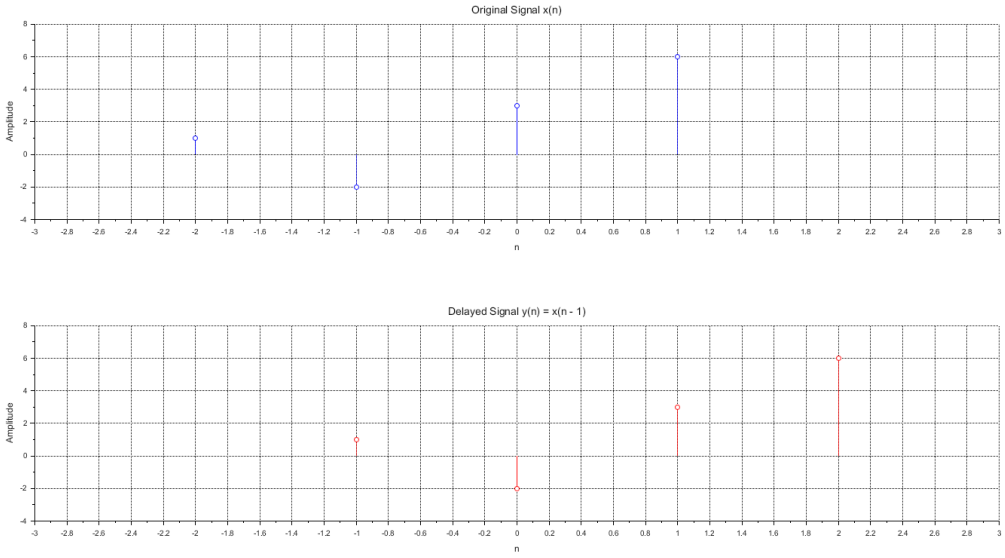


Figure 1: Original signal $x(n)$ and its delayed version $y(n) = x(n-1)$

4.2 Exercise 2 – Advance

Question: Write the function `[yn, yorigin] = advance(xn, xorigin, k)` that performs the advance operation $y(n) = x(n+k)$, where $k > 0$. $x(n)$ and $y(n)$ are graphically displayed in the same figure.

Explanation: Advancing a signal by k samples moves every sample k positions to the *left* on the time axis: the value that was at index $n+k$ is now at index n. The sample values themselves, and their order, do not change. What changes is which sample is at the origin. In the (`xn`, `xorigin`) form this means:

- The new origin is k positions further to the right inside the vector:

$$\texttt{yorigin} = \texttt{xorigin} + k. \tag{2}$$

- If `yorigin` is still inside the vector, `yn = xn`. Otherwise, the origin falls after the last sample, so zeros are appended on the right until the vector reaches position `yorigin`. This keeps the sample $y(0)$ (which is 0) inside the vector.

- The input $k \leq 0$ is rejected with `error()`, because the operation is only defined for $k > 0$.

For the example $x(n) = \{1, -2, \underset{\uparrow}{3}, 6\}$ (`xorigin = 3`) and $k = 1$:

$$y(n) = x(n+1) = \{1, -2, 3, \underset{\uparrow}{6}\}, \qquad \texttt{yn} = [1, -2, 3, 6], \quad \texttt{yorigin} = 4. \tag{3}$$

Check: $y(0) = x(1) = 6$ and $y(-3) = x(-2) = 1$. So the whole sequence now occupies $-3 \leq n \leq 0$, one sample to the left of the original $-2 \leq n \leq 1$.

Result:

Console output

```
yn =
1.    -2.    3.    6.
yorigin =
4.
```

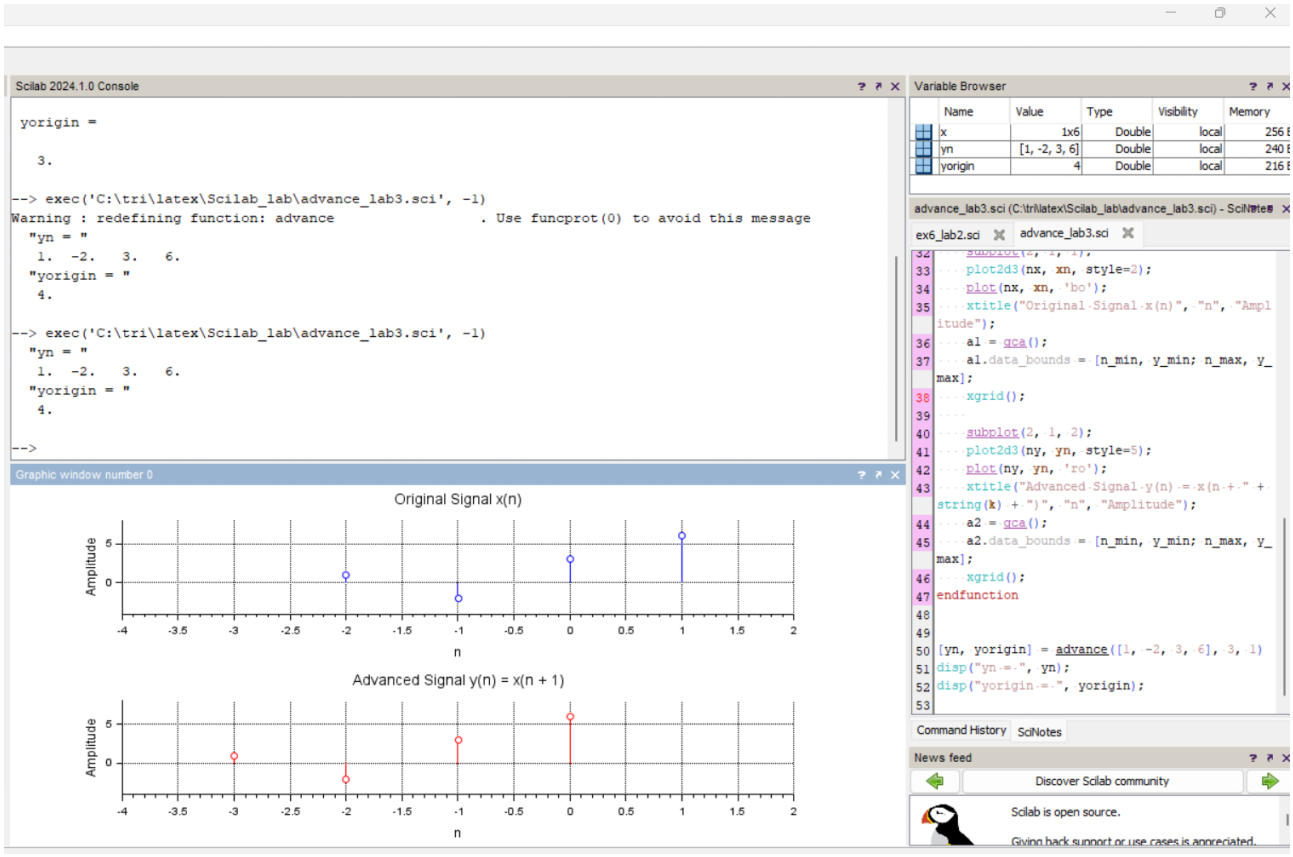


Figure 2: Result of Exercise 2 – `advance([1, -2, 3, 6], 3, 1)`: $x(n)$ (top) and $y(n) = x(n+1)$ (bottom)

The figure confirms the analysis: all four samples are shifted one unit to the left, and the sample 6 is now at $n = 0$.

4.3 Exercise 3 – Folding

Question: Write the function `[yn, yorigin] = fold(xn, xorigin)` that performs the folding operation $y(n) = x(-n)$. $x(n)$ and $y(n)$ are graphically displayed in the same figure.

Explanation:

Folding a discrete-time signal $x(n)$ to produce $y(n) = x(-n)$ means reflecting the signal across the vertical axis at $n = 0$. In an array-based representation, this operation is equivalent to reversing the order of the elements in the array.

For the time origin to remain aligned with $n = 0$, the origin index must also be transformed. If the original array has length L and the origin is at index `xorigin`, the new origin index

`yorigin` will be $L - \texttt{xorigin} + 1$.

For example, consider $x(n) = \{1, -2, \underset{\uparrow}{3}, 6\}$ with the origin at the 3rd element ($x(0) = 3$). Reversing the array gives $y(n) = \{6, \underset{\uparrow}{3}, -2, 1\}$. The new origin is calculated as $4 - 3 + 1 = 2$. Thus, the 2nd element of $y(n)$ is 3, confirming that $y(0) = x(0)$.

Result:

Console output

```
yn =
6.  3.  -2.  1.
yorigin =
2.
```

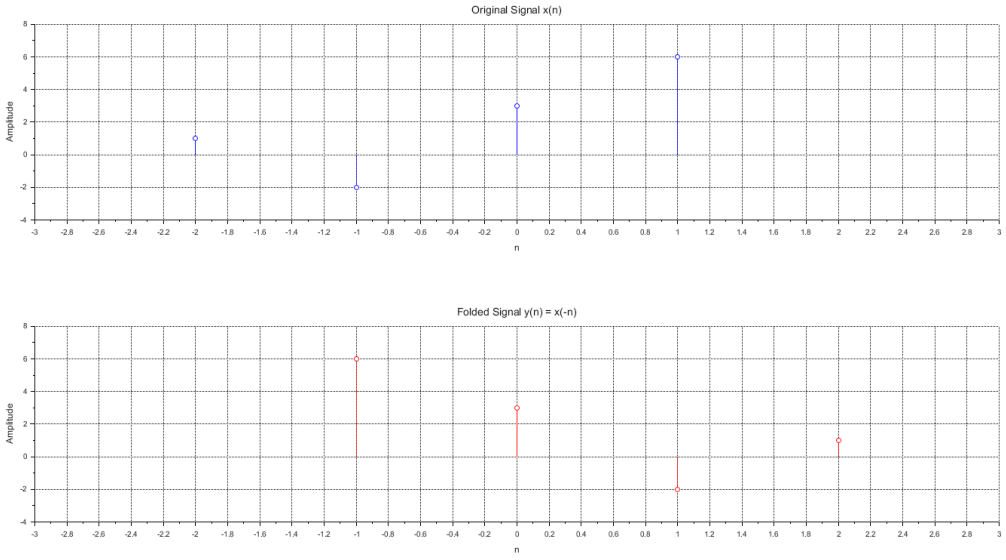


Figure 3: Original signal $x(n)$ and its folded version $y(n) = x(-n)$

4.4 Exercise 4 – Addition

Question: Write the function `[yn, yorigin] = add(x1n, x1origin, x2n, x2origin)` that performs the addition $y(n) = x_1(n) + x_2(n)$. $x_1(n)$, $x_2(n)$ and $y(n)$ are graphically displayed in the same figure.

For example, with $x_1(n) = \{\underset{\uparrow}{0}, 1, 3, -2\}$ and $x_2(n) = \{1, \underset{\uparrow}{1}, 2, 3\}$, the call `[yn, yorigin] = add([0, 1, 3, -2], 1, [1, 1, 2, 3], 2)` must return `yn = [1, 1, 3, 6, -2]` and `yorigin = 2`.

Explanation: Signals are added *sample by sample at the same time index*, not element by element in the vectors. If the two vectors have different lengths or different origins, they cannot be added directly with `+`. The function works in four steps:

1. Build the index vectors n_1 and n_2 of the two signals using (1).

2. Find the common time range that covers both signals:

$$n_{min} = \min(n_{1,min}, n_{2,min}), \qquad n_{max} = \max(n_{1,max}, n_{2,max}). \tag{4}$$

   The origin of the output is at position `yorigin` $= 1 - n_{min}$.

3. Zero-pad each signal on the left with $(n_{i,min} - n_{min})$ zeros and on the right with $(n_{max} - n_{i,max})$ zeros. After padding, both vectors have the same length and position i refers to the same time index in both.

4. Add the two padded vectors element-wise: `yn = x1_pad + x2_pad`.

For the example, $n_1 \in [0, 3]$ and $n_2 \in [-1, 2]$, so the common range is $[-1, 3]$:

n	-1	0	1	2	3
$x_1(n)$	0	0	1	3	-2
$x_2(n)$	1	1	2	3	0
$y(n)$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{3}$	$\mathbf{6}$	$\mathbf{-2}$

$$y(n) = \{1, \underset{\uparrow}{1}, 3, 6, -2\}, \qquad \texttt{yn} = [1, 1, 3, 6, -2], \quad \texttt{yorigin} = 1 - (-1) = 2, \tag{5}$$

which matches the expected output given in the assignment.

Result:

Console output

```
yn =
1.    1.    3.    6.    -2.
yorigin =
2.
```

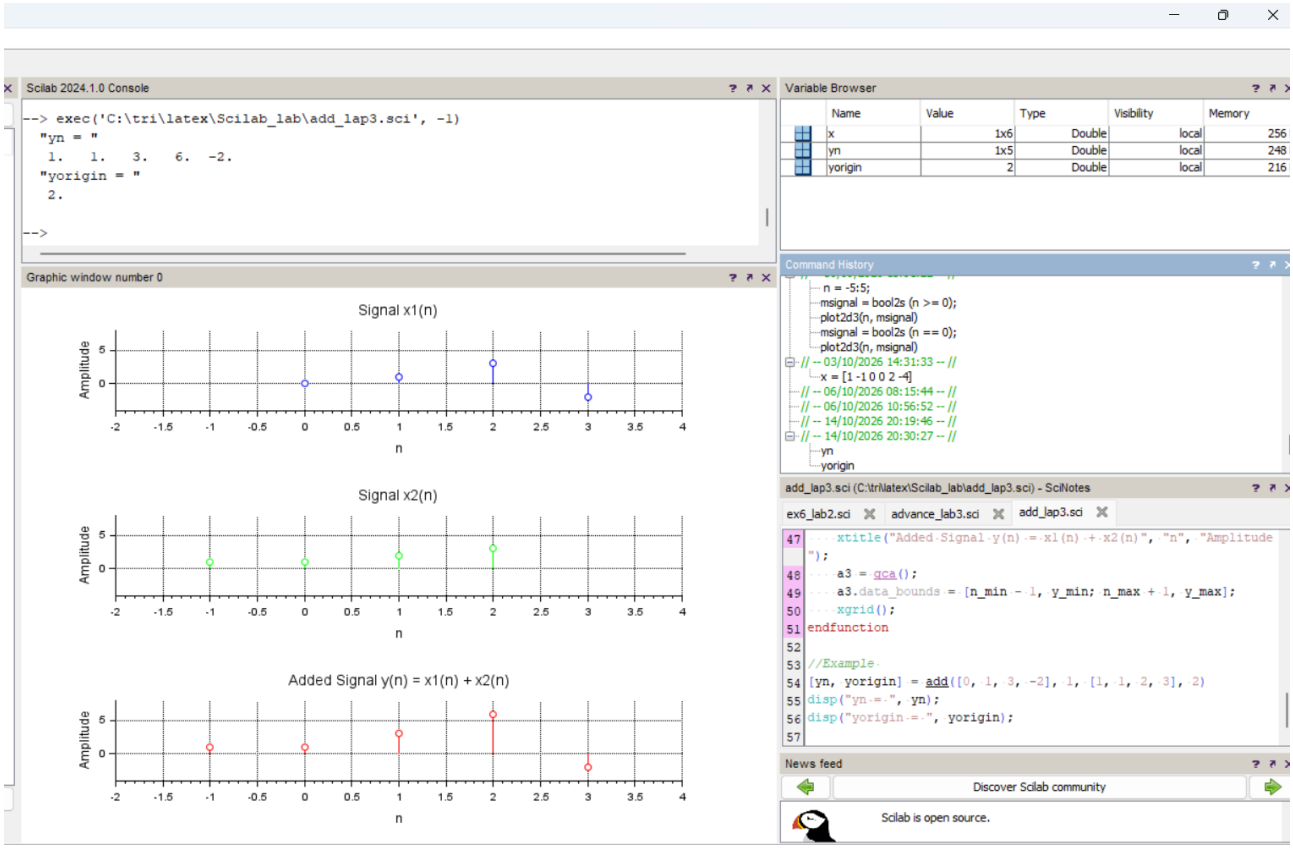


Figure 4: Result of Exercise 4 – `add([0, 1, 3, -2], 1, [1, 1, 2, 3], 2)`: $x_1(n)$, $x_2(n)$ and $y(n) = x_1(n) + x_2(n)$

4.5 Exercise 5 – Multiplication

Question: Write the function `[yn, yorigin] = multi(x1n, x1origin, x2n, x2origin)` that performs the multiplication $y(n) = x_1(n) \cdot x_2(n)$. $x_1(n)$, $x_2(n)$ and $y(n)$ are graphically displayed in the same figure.

Explanation:
To multiply two discrete-time signals, they must be evaluated over the same time index range. If the signals have different lengths or different origins, we must first align them by finding the overall minimum and maximum time indices. We then pad the signals with zeros outside their defined ranges so they share a common time vector. Multiplication is then performed element-by-element.

For example, let $x_1(n) = \{1, 2, \underset{\uparrow}{3}\}$ with the origin at the 3rd element ($x_1(0) = 3$), which spans $n \in [-2, 0]$. Let $x_2(n) = \{\underset{\uparrow}{4}, 5\}$ with the origin at the 1st element ($x_2(0) = 4$), which spans $n \in [0, 1]$. The common time span is $n \in [-2, 1]$. By zero-padding, we get aligned signals:

$$x_{1,\text{aligned}} = \{1, 2, \underset{\uparrow}{3}, 0\}, \qquad x_{2,\text{aligned}} = \{0, 0, \underset{\uparrow}{4}, 5\}$$

Multiplying these element-wise gives $y(n) = \{0, 0, \underset{\uparrow}{12}, 0\}$. The new origin corresponds to $n = 0$ in the common time vector, which is the 3rd element.

Result:

Console output

```
yn =
0.  0.  12.  0.
yorigin =
3.
```

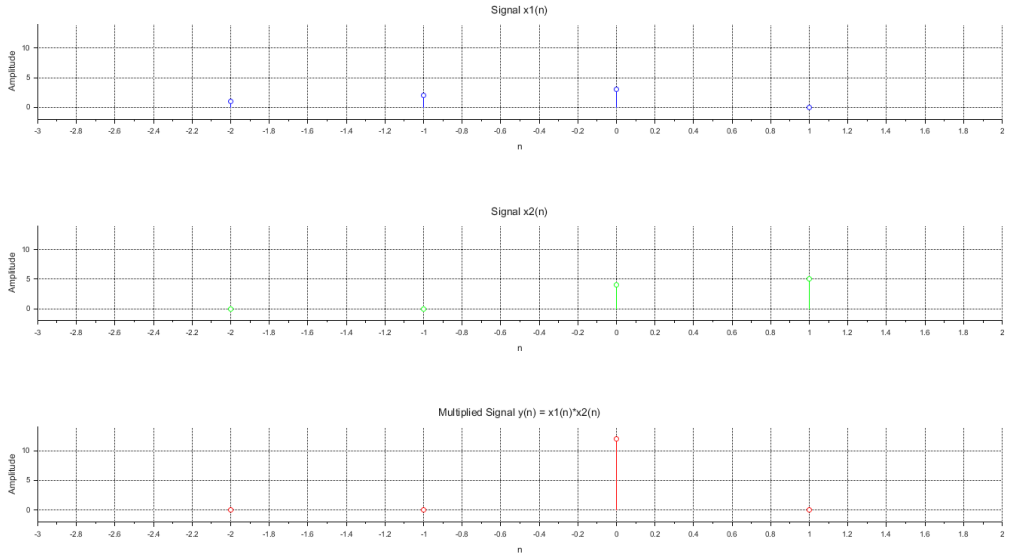


Figure 5: Signals $x_1(n)$, $x_2(n)$ and their element-wise product $y(n)$

4.6 Exercise 6 – Convolution

Question: Write the function `[yn, yorigin] = convolution(xn, xorigin, hn, horigin)` that performs the convolution $y(n) = x(n) * h(n)$, where $x(n)$ is the input signal and $h(n)$ is the impulse response of the system.

Explanation: The output of an LTI system is the convolution sum of the input and the impulse response:

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k). \tag{6}$$

- **Sample values.** For finite sequences, the convolution sum is the same as multiplying two polynomials. Scilab computes it directly with `conv(xn, hn)`. If $x(n)$ has N samples and $h(n)$ has M samples, $y(n)$ has $N + M - 1$ samples.

- **Origin.** The first sample of $y(n)$ is at the sum of the first indices of the two inputs:

$$n_{y,min} = n_{x,min} + n_{h,min} = (1 - \texttt{xorigin}) + (1 - \texttt{horigin}), \tag{7}$$

  so the origin of $y(n)$ is at position

$$\texttt{yorigin} = 1 - n_{y,min} = \texttt{xorigin} + \texttt{horigin} - 1. \tag{8}$$

Example: $x(n) = \{1, \underset{\uparrow}{2}, 3, 1\}$ (`xorigin = 2`) and $h(n) = \{\underset{\uparrow}{1}, 2, 1, -1\}$ (`horigin = 1`). The output starts at $n = -1 + 0 = -1$ and has $4 + 4 - 1 = 7$ samples. Computing the convolution sum by hand:

$$
\begin{aligned}
y(-1) &= x(-1)h(0) = 1 \\
y(0) &= x(-1)h(1) + x(0)h(0) = 2 + 2 = 4 \\
y(1) &= x(-1)h(2) + x(0)h(1) + x(1)h(0) = 1 + 4 + 3 = 8 \\
y(2) &= x(-1)h(3) + x(0)h(2) + x(1)h(1) + x(2)h(0) = -1 + 2 + 6 + 1 = 8 \\
y(3) &= x(0)h(3) + x(1)h(2) + x(2)h(1) = -2 + 3 + 2 = 3 \\
y(4) &= x(1)h(3) + x(2)h(2) = -3 + 1 = -2 \\
y(5) &= x(2)h(3) = -1
\end{aligned}
$$

$$
y(n) = \{1, \underset{\uparrow}{4}, 8, 8, 3, -2, -1\}, \qquad \texttt{yorigin} = 2 + 1 - 1 = 2. \tag{9}
$$

Result:

Console output

```
yn =
1.    4.    8.    8.    3.    -2.    -1.
yorigin =
2.
```

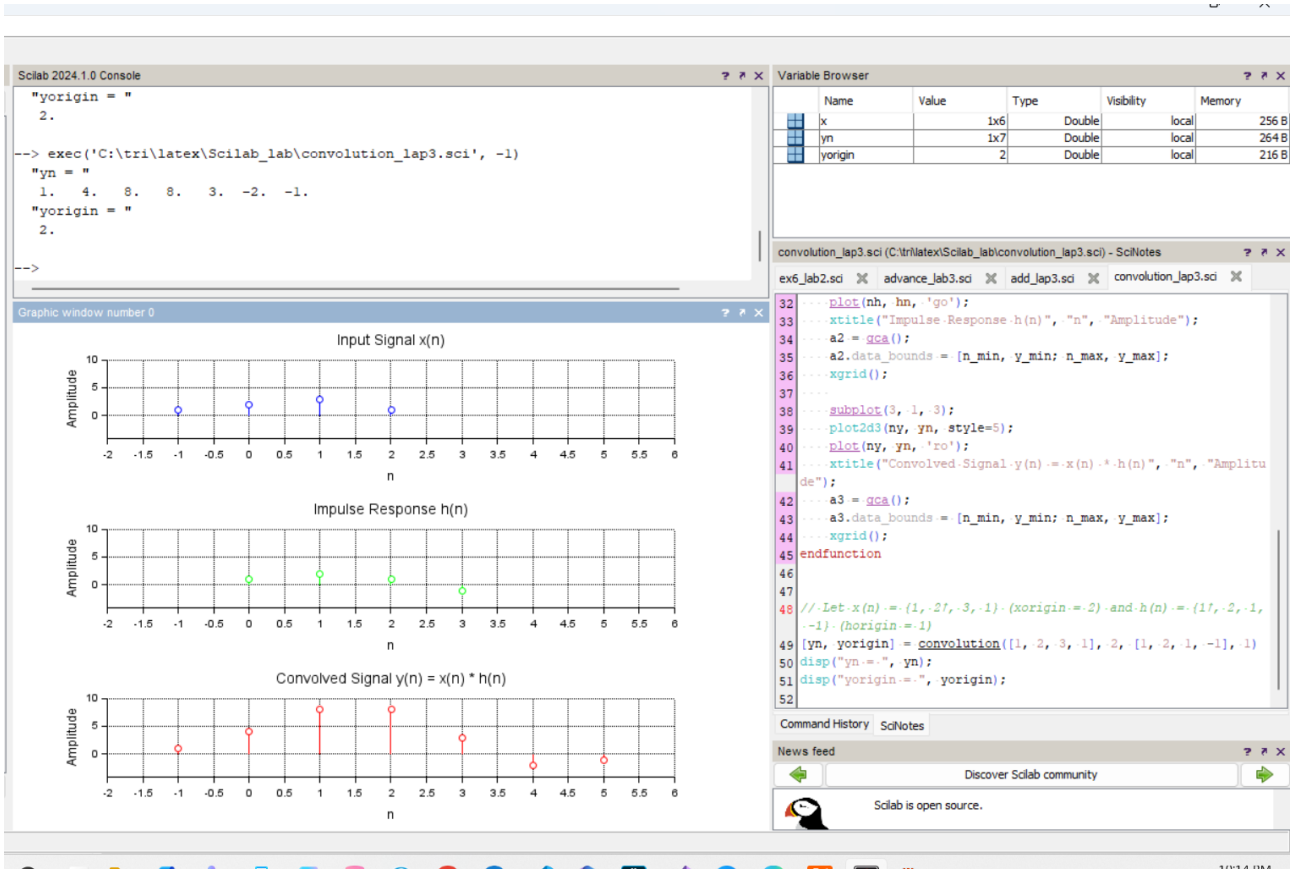


Figure 6: Result of Exercise 6 – `convolution([1, 2, 3, 1], 2, [1, 2, 1, -1], 1)`: $x(n)$, $h(n)$ and $y(n) = x(n) * h(n)$

The Scilab result matches the hand calculation exactly.

4.7 Additional Exercises

Additional exercise 2.1

A discrete-time signal $x(n)$ is defined as

$$x(n) = \begin{cases} 1 + \dfrac{n}{3}, & -3 \le n \le -1 \\ 1, & 0 \le n \le 3 \\ 0, & \text{elsewhere} \end{cases}$$

(a) Determine its values and sketch the signal $x(n)$.

(b) Sketch the signals that result if we:

1. First fold $x(n)$ and then delay the resulting signal by four samples.
2. First delay $x(n)$ by four samples and then fold the resulting signal.

(c) Sketch the signal $x(-n + 4)$.

(d) Compare the results in parts (b) and (c) and derive a rule for obtaining the signal $x(-n+k)$ from $x(n)$.

(e) Can you express the signal $x(n)$ in terms of signals $\delta(n)$ and $u(n)$?

Answer:

(a) The values of $x(n)$ evaluated at n for which it is nonzero are

$$x(n) = \left\{0, \frac{1}{3}, \frac{2}{3}, \underset{\uparrow}{1}, 1, 1, 1\right\}$$

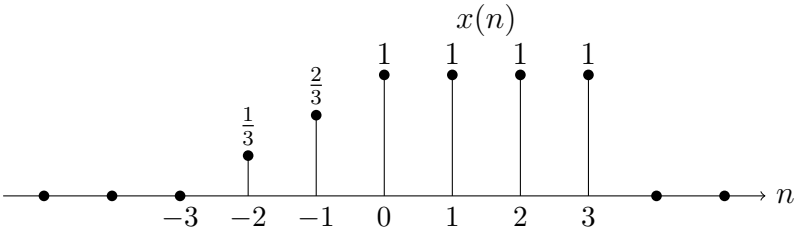


Figure 7: The signal $x(n)$

(b) 1. From folding $x(n)$ we have

$$x(-n) = \left\{1, 1, 1, \underset{\uparrow}{1}, \frac{2}{3}, \frac{1}{3}, 0\right\}$$

Delaying $x(-n)$ by four samples gives

$$x(-n+4) = \left\{\underset{\uparrow}{0}, 1, 1, 1, 1, \frac{2}{3}, \frac{1}{3}, 0\right\}$$

2. On the other hand, delaying $x(n)$ by 4 samples shifts the support from $[-3, 3]$ to $[1, 7]$:

$$x(n-4) = \left\{\underset{\uparrow}{0}, 0, \frac{1}{3}, \frac{2}{3}, 1, 1, 1, 1\right\}$$

Folding $x(n-4)$ (replacing n by $-n$) gives a signal with support $[-7, -1]$:

$$x(-n-4) = \left\{1, 1, 1, 1, \frac{2}{3}, \frac{1}{3}, 0, \underset{\uparrow}{0}\right\}$$

Check: $x(-n-4)$ at $n = -4$ equals $x(0) = 1$ and at $n = -2$ equals $x(-2) = \frac{1}{3}$.

(c) The signal $x(-n+4)$ is

$$x(-n+4) = \left\{\underset{\uparrow}{0}, 1, 1, 1, 1, \frac{2}{3}, \frac{1}{3}, 0\right\}$$

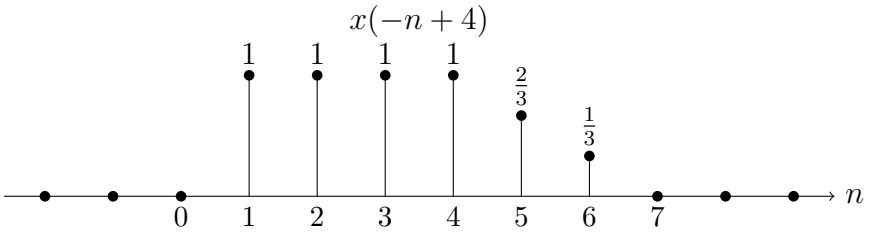


Figure 8: The signal $x(-n+4)$

(d) The result of (b.1) is identical to (c), while the result of (b.2), $x(-n-4)$, is different. Hence folding and time-shifting do *not* commute: the order of the two operations matters. Rule: to obtain $x(-n+k)$ from $x(n)$ (with $k>0$), first fold $x(n)$ to get $x(-n)$ and then delay the folded signal by k samples, since $x(-(n-k)) = x(-n+k)$. (Equivalently, first advance $x(n)$ by k samples to get $x(n+k)$ and then fold.)

(e) Any signal can be written as $x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$. Thus, we can write

$$\begin{aligned} x(n) &= \sum_{k=-2}^{3} x(k)\delta(n-k) \\ &= \frac{1}{3}\delta(n+2) + \frac{2}{3}\delta(n+1) + \delta(n) + \delta(n-1) + \delta(n-2) + \delta(n-3) \\ &= \frac{1}{3}\delta(n+2) + \frac{2}{3}\delta(n+1) + u(n) - u(n-4) \end{aligned}$$

Scilab: The signal $x(n)$ is written as a Scilab function `xsig(n)` that can be evaluated at *any* integer index vector. Each transformation is then obtained directly by changing the argument (e.g. `xsig(4 - n)` or `xsig(-4 - n)`) or applying element-wise operational logic over a common index grid $n = -8, \ldots, 8$.

Console output

```
(* marks the sample at n = 0; leading/trailing zeros are not printed)
x(n)                                    = {0.33, 0.67, 1*, 1, 1, 1},    n = -2..3
(b.1) fold x(n) then delay by 4:  x(-n+4)          = {0*, 1, 1, 1, 1, 0.67, 0.33},
n = 0..6
(b.2.i) delay x(n) by 4:  x(n-4)      = {0*, 0, 0.33, 0.67, 1, 1, 1, 1},    n = 0..7
(b.2.ii) fold x(n-4):  x(-n-4)      = {1, 1, 1, 1, 0.67, 0.33, 0, 0*},    n = -7..0
(c) x(-n+4)                             = {0*, 1, 1, 1, 1, 0.67, 0.33},    n = 0..6
(e) x(n) via delta & unit step          = {0.33, 0.67, 1*, 1, 1, 1},    n = -2..3
Check x(-n+4) from (b.1) == x(-n+4) from (c):  T
```

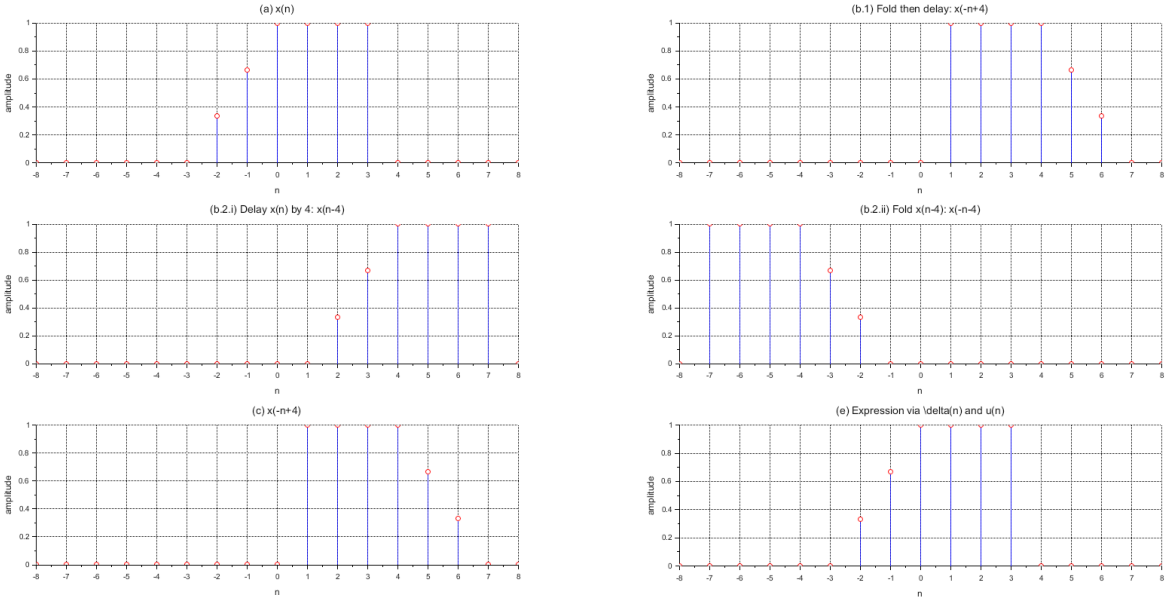


Figure 9: Discrete-time signal transformations and decomposition of $x(n)$

The Scilab results agree with all the hand-derived sequences in parts (a)–(e).

Additional exercise 2.2

A discrete-time signal $x(n)$ is shown in Fig. P2.2. Sketch and label carefully each of the following signals.

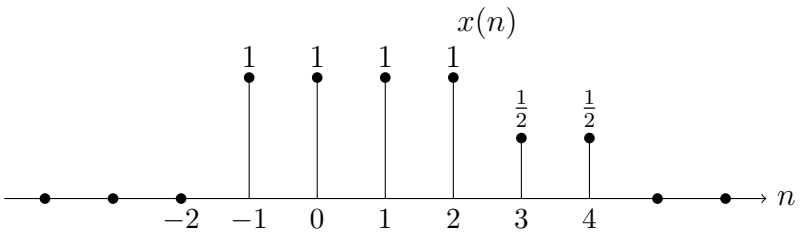


Figure 10: Figure P2.2 – the signal $x(n)$

(a) $x(n-2)$ **(b)** $x(4-n)$ **(c)** $x(n+2)$ **(d)** $x(n)u(2-n)$ **(e)** $x(n-1)\delta(n-3)$ **(f)** $x(n^2)$ **(g)** even part of $x(n)$ **(h)** odd part of $x(n)$

Answer: From the figure, $x(-1) = x(0) = x(1) = x(2) = 1$, $x(3) = x(4) = \frac{1}{2}$ and $x(n) = 0$ elsewhere:

$$x(n) = \{\ldots, 0, 1, \underset{\uparrow}{1}, 1, 1, \tfrac{1}{2}, \tfrac{1}{2}, 0, \ldots\}$$

(a) $x(n-2)$ is $x(n)$ *delayed* by 2 samples (shifted 2 units to the right). The support moves from $[-1, 4]$ to $[1, 6]$:

$$x(n-2) = \{\underset{\uparrow}{0}, 1, 1, 1, 1, \tfrac{1}{2}, \tfrac{1}{2}\}$$

(b) $x(4-n) = x\big(-(n-4)\big)$: first fold $x(n)$ to get $x(-n) = \{\frac{1}{2}, \frac{1}{2}, 1, 1, \underset{\uparrow}{1}, 1\}$ (support $[-4, 1]$), then delay the folded signal by 4 samples (support $[0, 5]$):

$$x(4-n) = \{\underset{\uparrow}{\tfrac{1}{2}}, \tfrac{1}{2}, 1, 1, 1, 1\}$$

Check: $y(0) = x(4) = \frac{1}{2}$ and $y(5) = x(-1) = 1$.

(c) $x(n+2)$ is $x(n)$ *advanced* by 2 samples (shifted 2 units to the left). The support moves to $[-3, 2]$:

$$x(n+2) = \{1, 1, 1, \underset{\uparrow}{1}, \tfrac{1}{2}, \tfrac{1}{2}\}$$

(d) $u(2-n) = 1$ for $n \le 2$ and 0 for $n > 2$, so it keeps only the samples with $n \le 2$:

$$x(n)u(2-n) = \{1, \underset{\uparrow}{1}, 1, 1\}$$

(e) $\delta(n-3)$ is non-zero only at $n = 3$, so $x(n-1)\delta(n-3) = x(3-1)\,\delta(n-3) = x(2)\,\delta(n-3) = \delta(n-3)$:

$$x(n-1)\delta(n-3) = \{\underset{\uparrow}{0}, 0, 0, 1\}$$

(f) $x(n^2)$: $n = 0 \Rightarrow x(0) = 1$; $n = \pm 1 \Rightarrow x(1) = 1$; $n = \pm 2 \Rightarrow x(4) = \frac{1}{2}$; $|n| \ge 3 \Rightarrow n^2 \ge 9 \Rightarrow x(n^2) = 0$:

$$x(n^2) = \{\tfrac{1}{2}, 1, \underset{\uparrow}{1}, 1, \tfrac{1}{2}\}$$

(g)(h) The even and odd parts are $x_e(n) = \frac{x(n)+x(-n)}{2}$ and $x_o(n) = \frac{x(n)-x(-n)}{2}$:

n	-4	-3	-2	-1	0	1	2	3	4
$x(n)$	0	0	0	1	1	1	1	$\frac{1}{2}$	$\frac{1}{2}$
$x(-n)$	$\frac{1}{2}$	$\frac{1}{2}$	1	1	1	1	0	0	0
$x_e(n)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$	1	1	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
$x_o(n)$	$-\frac{1}{4}$	$-\frac{1}{4}$	$-\frac{1}{2}$	0	0	0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

$$x_e(n) = \{\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{2}, 1, \underset{\uparrow}{1}, 1, \tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}\}, \qquad x_o(n) = \{-\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{2}, 0, \underset{\uparrow}{0}, 0, \tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}\}$$

We can check that $x_e(n) = x_e(-n)$, $x_o(n) = -x_o(-n)$ and $x_e(n) + x_o(n) = x(n)$.

Scilab: The signal $x(n)$ is written as a Scilab function `xsig(n)` that can be evaluated at *any* integer index vector. Each transformation is then obtained just by changing the argument, e.g. `xsig(4 - n)` or `xsig(n.*2)`, over a common index grid $n = -6, \ldots, 8$.

Console output

```
(* marks the sample at n = 0)
x(n)                                  = {1, 1*, 1, 1, 0.5, 0.5},     n = -1..4
(a) x(n-2)                          = {0*, 1, 1, 1, 1, 0.5, 0.5},    n = 0..6
(b) x(4-n)                            = {0.5*, 0.5, 1, 1, 1, 1},     n = 0..5
(c) x(n+2)                          = {1, 1, 1, 1*, 0.5, 0.5},       n = -3..2
```

```
(d) x(n)u(2-n)                                          = {1, 1*, 1, 1},     n = -1..2
(e) x(n-1)d(n-3)                                          = {0*, 0, 0, 1},     n = 0..3
(f) x(n^2)                                        = {0.5, 1, 1*, 1, 0.5},     n = -2..2
(g) xe(n)              = {0.25, 0.25, 0.5, 1, 1*, 1, 0.5, 0.25, 0.25},     n = -4..4
(h) xo(n)          = {-0.25, -0.25, -0.5, 0, 0*, 0, 0.5, 0.25, 0.25},     n = -4..4
Check xe(n) + xo(n) = x(n):  T
```

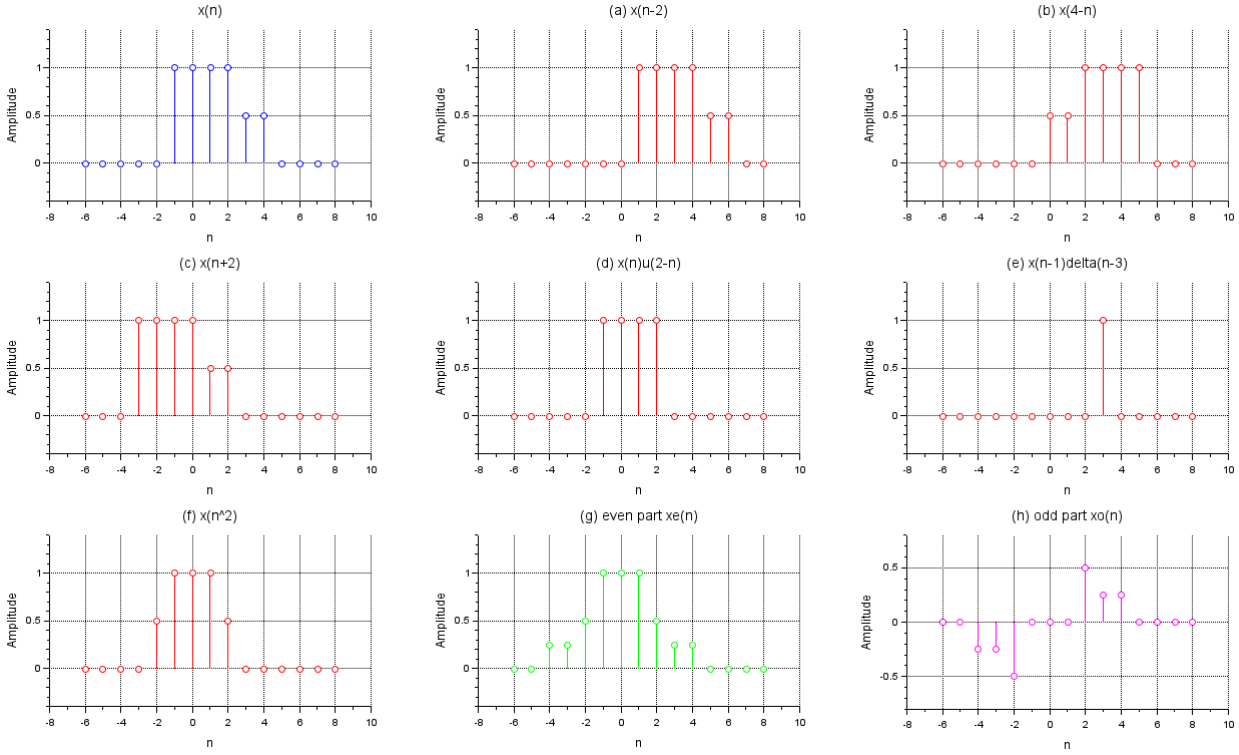


Figure 11: Additional exercise 2.2 – $x(n)$ and the signals (a)–(h)

The Scilab results agree with all the hand-derived sequences (a)–(h).

Additional exercise 2.4

Show that any signal can be decomposed into an even and an odd component. Is the decomposition unique? Illustrate your arguments using the signal

$$x(n) = \{2, 3, \underset{\uparrow}{4}, 5, 6\}$$

Answer:
Existence. For any signal $x(n)$, define

$$x_e(n) = \frac{1}{2}\big[x(n) + x(-n)\big], \qquad x_o(n) = \frac{1}{2}\big[x(n) - x(-n)\big]. \tag{10}$$

Then

- $x_e(-n) = \frac{1}{2}\big[x(-n) + x(n)\big] = x_e(n)$, so $x_e(n)$ is **even**;

- $x_o(-n) = \frac{1}{2}\big[x(-n) - x(n)\big] = -x_o(n)$, so $x_o(n)$ is **odd**;
- $x_e(n) + x_o(n) = \frac{1}{2}x(n) + \frac{1}{2}x(-n) + \frac{1}{2}x(n) - \frac{1}{2}x(-n) = x(n)$.

Hence every signal can be written as the sum of an even and an odd component.
Uniqueness. Suppose $x(n) = e(n) + o(n)$ for *some* even signal $e(n)$ and *some* odd signal $o(n)$. Replacing n by $-n$ and using $e(-n) = e(n)$, $o(-n) = -o(n)$:

$$\begin{cases} x(n) = e(n) + o(n) \\ x(-n) = e(n) - o(n) \end{cases} \implies e(n) = \frac{x(n) + x(-n)}{2} = x_e(n), \quad o(n) = \frac{x(n) - x(-n)}{2} = x_o(n).$$

Any even/odd pair that adds up to $x(n)$ must equal $x_e(n)$ and $x_o(n)$, so **the decomposition is unique**.
Illustration. With $x(n) = \{2, 3, \underset{\uparrow}{4}, 5, 6\}$ we have $x(-n) = \{6, 5, \underset{\uparrow}{4}, 3, 2\}$:

n	-2	-1	0	1	2
$x(n)$	2	3	4	5	6
$x(-n)$	6	5	4	3	2
$x_e(n)$	4	4	4	4	4
$x_o(n)$	-2	-1	0	1	2

$$x_e(n) = \{4, 4, \underset{\uparrow}{4}, 4, 4\}, \qquad x_o(n) = \{-2, -1, \underset{\uparrow}{0}, 1, 2\}$$

$x_e(n)$ is symmetric about $n = 0$, $x_o(n)$ is anti-symmetric with $x_o(0) = 0$, and their sum gives back $\{2, 3, \underset{\uparrow}{4}, 5, 6\}$. For example, the split $\{2, 3, \underset{\uparrow}{4}, 5, 6\} = \{2, 3, \underset{\uparrow}{4}, 3, 2\} + \{0, 0, \underset{\uparrow}{0}, 2, 4\}$ also sums to $x(n)$, but the second part is not odd. By the uniqueness proof, no other even/odd pair is possible.
Scilab: The function `evenodd(xn, xorigin)` uses the same (vector, origin) convention as the lab functions. It places $x(n)$ on a symmetric grid $[-L, L]$, so folding is just reversing the vector (`xs($:-1:1)`).

Console output

```
n       = -2   -1   0   1   2
x(n)     = 2   3   4   5   6
x(-n)   = 6   5   4   3   2
xe(n)   = 4   4   4   4   4
xo(n)   = -2   -1   0   1   2
yorigin = 3
xe even (xe(n) = xe(-n)) :  T
xo odd (xo(n) = -xo(-n)) :  T
xe + xo == x             :  T
```

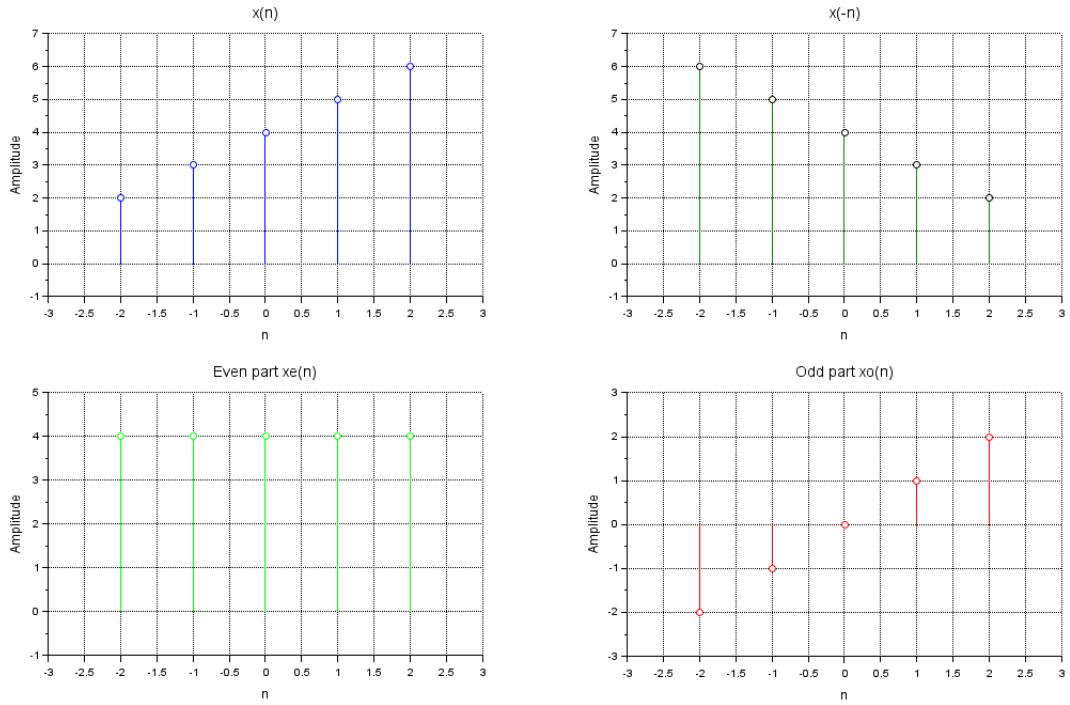


Figure 12: Additional exercise 2.4 – $x(n)$, $x(-n)$, even part $x_e(n)$ and odd part $x_o(n)$

Additional exercise 2.6

Consider the system $y(n) = \mathcal{T}[x(n)] = x(n^2)$.

(a) Determine if the system is time invariant.

(b) To clarify the result in part (a) assume that the signal $x(n) = \begin{cases} 1, & 0 \le n \le 3 \\ 0, & \text{elsewhere} \end{cases}$ is applied into the system. (1) Sketch the signal $x(n)$. (2) Determine and sketch the signal $y(n) = \mathcal{T}[x(n)]$. (3) Sketch the signal $y_2'(n) = y(n-2)$. (4) Determine and sketch the signal $x_2(n) = x(n-2)$. (5) Determine and sketch the signal $y_2(n) = \mathcal{T}[x_2(n)]$. (6) Compare the signals $y_2(n)$ and $y(n-2)$. What is your conclusion?

(c) Repeat part (b) for the system $y(n) = x(n) - x(n-1)$. Can you use this result to make any statement about the time invariance of this system? Why?

(d) Repeat parts (b) and (c) for the system $y(n) = \mathcal{T}[x(n)] = n\,x(n)$.

Answer:

A system is **time invariant** if shifting the input by k only shifts the output by the same k:

$$x(n) \xrightarrow{\mathcal{T}} y(n) \quad \Longrightarrow \quad x(n-k) \xrightarrow{\mathcal{T}} y(n-k), \quad \forall\, x(n), \forall\, k.$$

(a) For $y(n) = x(n^2)$, the response to the delayed input $x(n-k)$ is obtained by replacing n by n^2 *inside the argument of the input only*:

$$y(n,k) = \mathcal{T}[x(n-k)] = x(n^2-k),$$

while the delayed output is

$$y(n-k) = x\big((n-k)^2\big) = x(n^2 - 2nk + k^2).$$

Since $x(n^2-k) \neq x\big((n-k)^2\big)$ in general, the system is **time variant**.

(b) With $x(n) = \{\underset{\uparrow}{1}, 1, 1, 1\}$:

(1) $x(n) = \{\underset{\uparrow}{1}, 1, 1, 1\}$, $0 \le n \le 3$.

(2) $y(n) = x(n^2)$: $y(0) = x(0) = 1$, $y(\pm 1) = x(1) = 1$, $y(\pm 2) = x(4) = 0$, so

$$y(n) = \{1, \underset{\uparrow}{1}, 1\}$$

(3) $y_2'(n) = y(n-2) = \{\underset{\uparrow}{0}, 1, 1, 1\}$ (non-zero for $1 \le n \le 3$).

(4) $x_2(n) = x(n-2) = \{\underset{\uparrow}{0}, 0, 1, 1, 1, 1\}$ (non-zero for $2 \le n \le 5$).

(5) $y_2(n) = x_2(n^2) = x(n^2-2)$, which equals 1 only when $0 \le n^2 - 2 \le 3 \Leftrightarrow 2 \le n^2 \le 5 \Leftrightarrow n = \pm 2$:

$$y_2(n) = \{1, 0, \underset{\uparrow}{0}, 0, 1\}$$

(6) $y_2(n) = \{1, 0, \underset{\uparrow}{0}, 0, 1\} \neq y(n-2) = \{\underset{\uparrow}{0}, 1, 1, 1\}$. One counter-example is enough to show that the system is **time variant**, which confirms part (a).

(c) System $y(n) = x(n) - x(n-1)$:

(1) $x(n) = \{\underset{\uparrow}{1}, 1, 1, 1\}$.

(2) $y(n) = x(n) - x(n-1) = \{\underset{\uparrow}{1}, 0, 0, 0, -1\}$ ($y(0) = 1 - 0$, $y(4) = 0 - 1$).

(3) $y(n-2) = \{\underset{\uparrow}{0}, 0, 1, 0, 0, 0, -1\}$ (non-zero at $n = 2$ and $n = 6$).

(4) $x_2(n) = x(n-2) = \{\underset{\uparrow}{0}, 0, 1, 1, 1, 1\}$.

(5) $y_2(n) = x_2(n) - x_2(n-1) = \{\underset{\uparrow}{0}, 0, 1, 0, 0, 0, -1\}$.

(6) $y_2(n) = y(n-2)$ for this input.

Can we conclude time invariance from this result? *No.* Time invariance must hold for *every* input and *every* shift k. One example where $y_2(n) = y(n-2)$ is consistent with time invariance but does not prove it. (A single counter-example, on the other hand, is enough to prove time *variance*.) To prove it we need the general argument:

$$\mathcal{T}[x(n-k)] = x(n-k) - x(n-k-1) = y(n-k) \quad \forall\, x, k,$$

so the system $y(n) = x(n) - x(n-1)$ is indeed **time invariant**.

(d) System $y(n) = n\,x(n)$:

(1) $x(n) = \{\underset{\uparrow}{1}, 1, 1, 1\}$.

(2) $y(n) = n\,x(n) = \{\underset{\uparrow}{0}, 1, 2, 3\}$.

(3) $y(n-2) = (n-2)\,x(n-2) = \{\underset{\uparrow}{0}, 0, 0, 1, 2, 3\}$ (values $0, 1, 2, 3$ at $n = 2, \ldots, 5$).

(4) $x_2(n) = x(n-2) = \{\underset{\uparrow}{0}, 0, 1, 1, 1, 1\}$.

(5) $y_2(n) = n\,x_2(n) = \{\underset{\uparrow}{0}, 0, 2, 3, 4, 5\}$.

(6) $y_2(n) \neq y(n-2)$ (e.g. $y_2(2) = 2$ but $y(0) = 0$), so the system is **time variant**.

Statement about time invariance (part (c) for this system): here the example *is* enough to decide, because one counter-example proves time variance. In general,

$$\mathcal{T}[x(n-k)] = n\,x(n-k) \;\neq\; y(n-k) = (n-k)\,x(n-k).$$

The gain n depends explicitly on time, so the system is time variant.

Table 2: Summary of Additional exercise 2.6

System	$y_2(n)$ vs. $y(n-2)$	Conclusion
$y(n) = x(n^2)$	different	Time variant
$y(n) = x(n) - x(n-1)$	equal	Time invariant (proved analytically)
$y(n) = n\,x(n)$	different	Time variant

Scilab: Each system is written as a function `T(f, n)` that receives the *input signal as a function* `f` and returns the output at the indices `n`. This way, $y(n-2)$ is simply `T(xin, n-2)` and $y_2(n) = \mathcal{T}[x(n-2)]$ is `T(x2in, n)`. The function `check_system` performs steps (1)–(6) for each system. Step (6) is plotted as the difference $y_2(n) - y(n-2)$: if it is identically zero, the shift of the input only shifts the output.

Console output

```
(* marks the sample at n = 0)
--- System 1:  y(n) = x(n^2) ---
x(n)        = {1*, 1, 1, 1},    n = 0..3
y(n)        = {1, 1*, 1},    n = -1..1
y(n-2)      = {0*, 1, 1, 1},    n = 0..3
x2(n)=x(n-2) = {0*, 0, 1, 1, 1, 1},    n = 0..5
y2(n)=T[x2]  = {1, 0, 0*, 0, 1},    n = -2..2
⇒ y2(n) ≠ y(n-2) ⟹ the system is TIME-VARIANT
--- System 2:  y(n) = x(n) - x(n-1) ---
x(n)        = {1*, 1, 1, 1},    n = 0..3
y(n)        = {1*, 0, 0, 0, -1},    n = 0..4
y(n-2)      = {0*, 0, 1, 0, 0, 0, -1},    n = 0..6
x2(n)=x(n-2) = {0*, 0, 1, 1, 1, 1},    n = 0..5
y2(n)=T[x2]  = {0*, 0, 1, 0, 0, 0, -1},    n = 0..6
⇒ y2(n) = y(n-2) for this input (consistent with time-invariance)
--- System 3:  y(n) = n x(n) ---
```

```
x(n)        = {1*, 1, 1, 1},    n = 0..3
y(n)        = {0*, 1, 2, 3},    n = 0..3
y(n-2)      = {0*, 0, 0, 1, 2, 3},    n = 0..5
x2(n)=x(n-2) = {0*, 0, 1, 1, 1, 1},    n = 0..5
y2(n)=T[x2]  = {0*, 0, 2, 3, 4, 5},    n = 0..5
⇒ y2(n) ≠ y(n-2) ⟹ the system is TIME-VARIANT
```

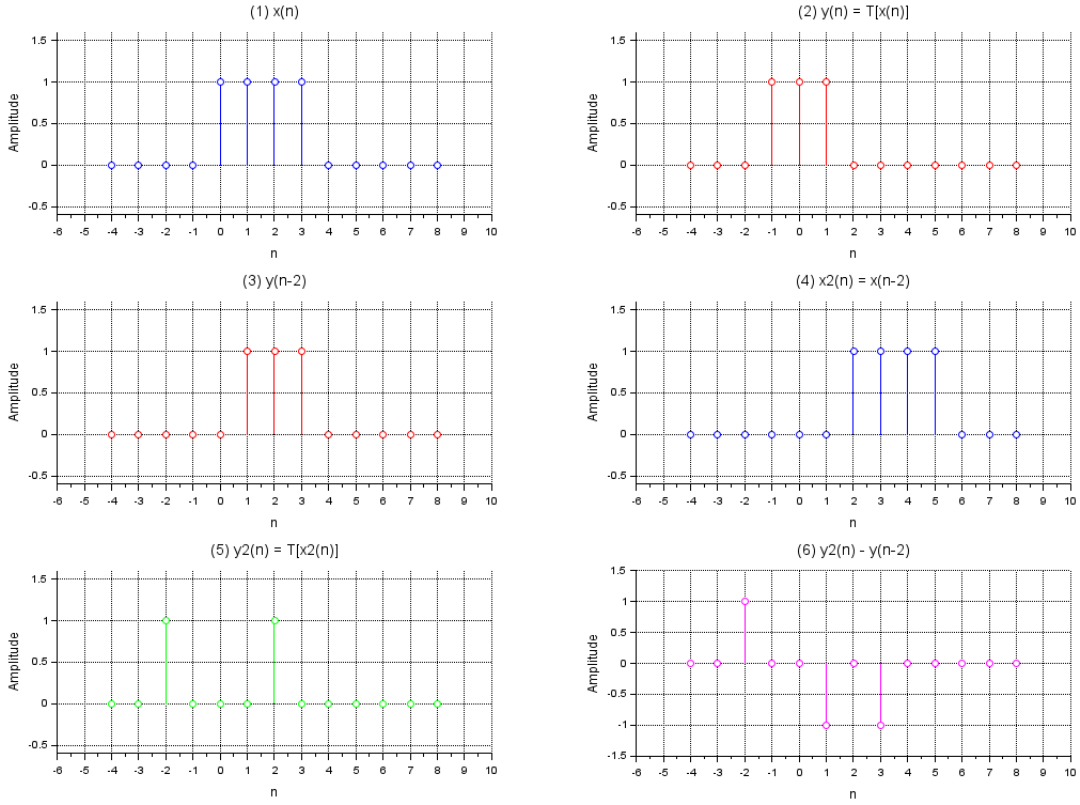


Figure 13: Additional exercise 2.6(b) – system $y(n) = x(n^2)$: $y_2(n) - y(n-2) \neq 0 \Rightarrow$ time variant

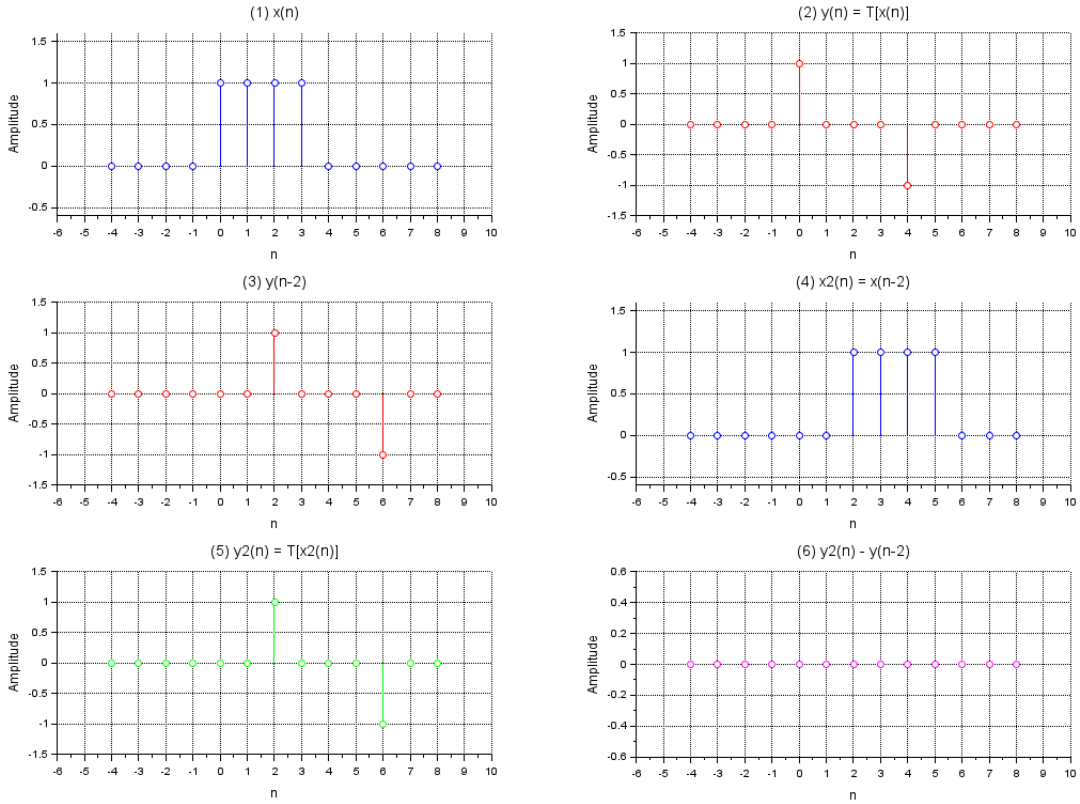


Figure 14: Additional exercise 2.6(c) – system $y(n) = x(n) - x(n-1)$: $y_2(n) - y(n-2) = 0$

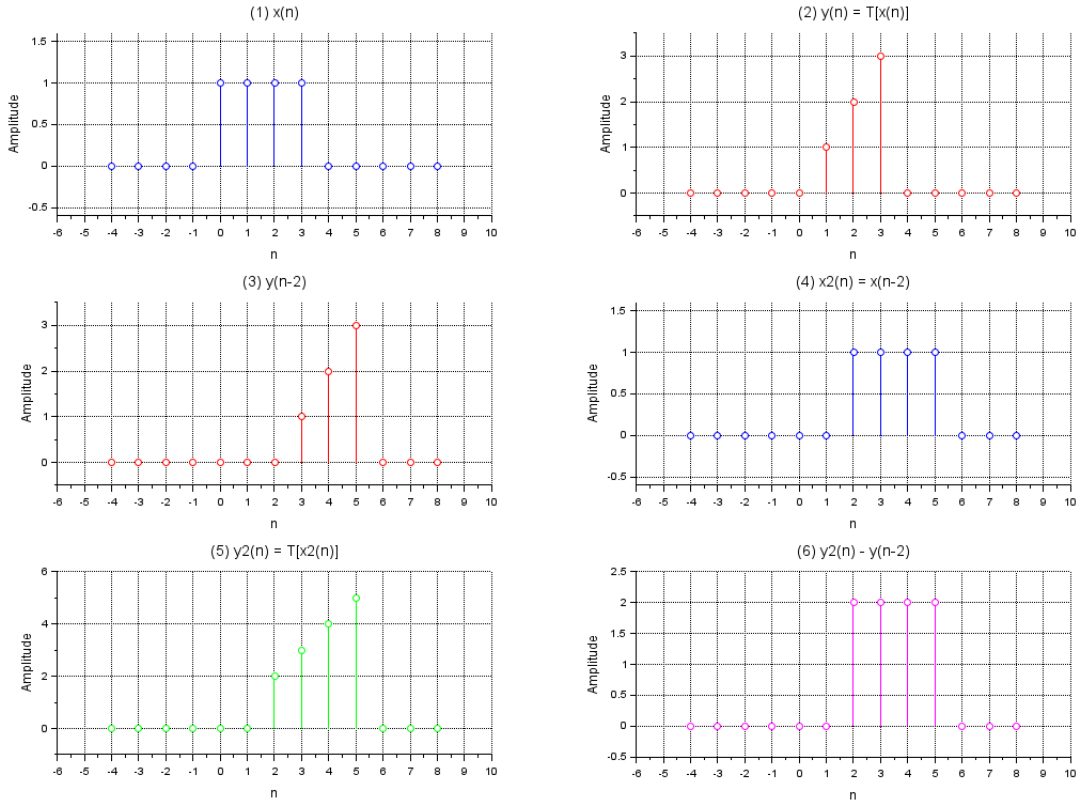


Figure 15: Additional exercise 2.6(d) – system $y(n) = n\,x(n)$: $y_2(n) - y(n-2) \neq 0 \Rightarrow$ time variant

5 Discussion

Representing a signal by the pair (vector, origin) made it possible to implement every time-domain operation as a general Scilab function. The exercises show that most operations only need careful index bookkeeping:

- Time shifts (`advance`, `delay`) leave the sample vector unchanged and only move the origin. Zero-padding is needed only when the origin falls outside the vector.
- Sample-by-sample operations (`add`, `multi`) require both signals to be aligned on a common index range by zero-padding before using the element-wise operators.
- For `convolution`, the built-in `conv` gives the sample values, and the origin follows from $n_{y,min} = n_{x,min} + n_{h,min}$.

In the additional exercises, evaluating a signal as a function of an arbitrary index vector turned out to be a convenient way to compute folded, shifted or non-linearly indexed signals such as $x(n^2)$. Exercise 2.6 also shows that one example can only *disprove* time invariance. Proving it requires a general argument.

6 References

- J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th ed., Pearson, 2007 – Chapter 2 Problems.
- Digital Signal Processing Lecture Notes - Ho Chi Minh City University of Technology.
- Scilab Online Manuals: https://help.scilab.org/doc/5.5.2/en_US/function.html, https://help.scilab.org/doc/5.5.2/en_US/deff.html.

7 Source Code

The source code, scripts, and LaTeX files for this laboratory report can be accessed at our GitHub repository:
https://github.com/aulequangtri/signal_scilab